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9) µC vs µp

A) µC :-

→ External processor along with internal memory and I/O components
→ Circuit is very small since I/O comps are less
→ Cost of system is low and compact
→ Power consumption is ~~if~~ ~~fast~~ less
→ Based on harvard architecture, program and data memory are seperate

Microprocessor :-

→ It is just a processor, memory and I/O have to be connected seperately
→ Circuit is large since I/O is external
→ Dont have power saving, not ~~lower~~ low power or compact
→ Have less registers so more operations are memory based
→ Based on von neuman architecture where program and data in same module.

Q) Diff b/w RISC and CISC

Risc :-

→ Uses small simple fixed length instructions and follow load/store approach enabling efficient and fast execution
→ Instruction come in form of one word
→ Single clock cycle to execute
→ More general purpose registers, register - register operation
→ Simple addressing
→ optimised pipeline due to uniform instruction size and complexity

CISC :

→ Uses a large set of complex, variable length instructions that can perform multiple operations, often requiring multiple clock cycles.

→ Instructions are larger than one - word size.

→ Instructions take more than one cycle

→ Less number of general purpose registers

→ Complex addressing mode

Q) ARM family duplex. And if simplex to duplex

A) It depends on the chip and board architecture, the ARM4 series supports ~~both simplex and~~ half duplex. Its not fully simplex or duplex. It can be used in duplex mode by configuring one pin as input, other gpio pin as output thereby obtaining functionality of full duplex communication.

Q) Advantages

RISC over CISC—

→ Low power design

→ Execution in single instruction cycle

→ Better pipeling can overlap tasks

→ Smaller chip size and lower cost

CISC over RISC :

→ Higher code density

→ Less compact complex tasks using fewer instructions

→ DMA, direct memory access registers are not required

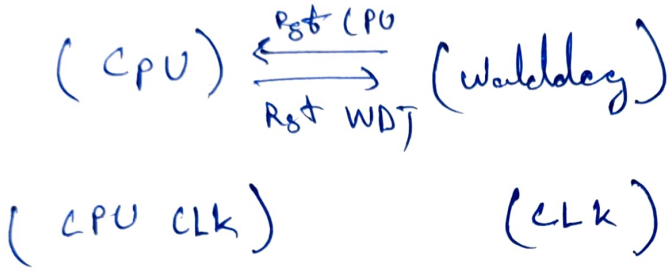
→ Maintain support for legacy software i.e backwards compatibility.

5) Watchdog timer Vs Flag bit

Watchdog timer :

→ It's a hardware timer that automatically generates a system reset if the main program neglects to periodically service it. Resets an embedded system that crashes or loops.

→ Uses an internal 128 kHz clock source and needs to be configured. When enabled counts from 0 to a user defined value.



→ Timer is loaded with an initial value greater than the worst-case time delay through the main program loop, so each time it goes through the main loop the code resets the watchdog timer.

Flag timer :

→ Single bit indicators stored inside a special register (Eg: program status word or special register)

→ They are automatically set (1) or cleared (0) by the hardware to show the state or result of a math, logic or hardware event, helping the CPU make choices

→ Common types : Carry flag (CY), Zero flag (Z), Overflow flag (OV), Parity flag (P), Auxiliary Carry (AC)

→ Peripheral and Interrupt flags : Timer / Counter overflow flag, UART (shows if new serial data available), ADC conversion complete flag.